

JPE Replication Report

PUBLISHED
June 5, 2026

1 Paper Identification

ID	20240098
Author	Sollaci
Title	Managers and Productivity in Retail
Iteration	2

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. There is a missing R package from your readme.
2. We were able to fix a few bugs. Please implement those fixes.
3. In your Data Availability Section in the readme, delete the second paragraph.
4. There are a few discrepancies in output that you should check.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☐ Details on each Data Source
- ☒ Dataset list

- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory
 - ☒ Runtime
 - ☐ and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
- ☐ {REPLICATOR} to Replicators
 - ☐ Details (as necessary)
- ☐ List of tables and programs
- ☐ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☐ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☒ is a simulation study: the data are generated by an algorithm. (Parts of it.)
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

The data was provided to the authors under the condition that the companies the data stems from remain anonymous. The data was provided for replication purposes but data sources, references and access conditions were not provided for this reason.

The authors specified that the data "should be preserved in the restricted-access location of the Journal of Political Economy for as long as necessary for the successful replication of the paper. It

should be deleted after the replication is confirmed. The authors commit to preserving the data and code for a period of at least five years following the publication of the paper and to provide reasonable assistance to requests for clarification and replication."

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README:

- The authors provided a codebook for the data describing the variables contained in the datasets.

5.1 Data Sources

file/name	public	private	DOI/URL	Access		
				described	cited readme	cited paper
data/A_dataset.dta	no	yes	no	no	no	no
data/B_dataset.dta	no	yes	no	no	no	no

5.1.1 Data

Data are not cited, no access description provided either, no information regarding the data is provided.

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided (not for publication)
- ☐ DOI or URL is provided, and works: <http://data.bls.gov/cgi-bin/surveymost?sm+08>
- ☐ Access conditions are described:

☐ No information on access conditions.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☐ Data are cited

☐ in the manuscript

☐ in the README:

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - R (version 4.4.0)
 - Stata (version 19)
 - Python (version 3.11.11)
- ☒ Computational Requirements specified as follows:
 - R: packages: haven, rstudioapi, doBy, lfe, Matrix
 - Stata: packages to be installed: reghdfe, ftools, did_imputation, event_plot, binscatter, grc1leg2, white_tableau scheme
 - Python: libraries: numpy, pandas, scipy, matplotlib, os
 - ebayes.ado and fese_fast.ado (provided in repository)
- ☒ Time Requirements specified as follows:
 - ca. 2h
- ☐ Requirements are complete.

For missing requirements, see the list of required changes in [Section 9](#)

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 2 - Variables flagged in data: 24 - Code files with PII references: 20 - PII references in code: 525

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	A_dataset.dta	4	loc, location, name
Data	B_dataset.dta	20	name, lon, lat, country, minute, loc, location, city
Code	A_estimate_fe.R	12	name, network, lat
Code	A_estimate_gfe.R	12	name, network, lat

File Type	File	Variables/References	PII Categories
Code	A_event_study.do	64	loc, name, lon, son, lat
Code	A_fixed_effects.do	10	name, lat, loc, location, lon
Code	A_summary_stats.do	10	loc, location, lat, name
Code	B_estimate_fe.R	12	name, network, lat
Code	B_estimate_gfe.R	12	name, network, lat
Code	B_event_study.do	64	loc, name, lon, son, lat
Code	B_fixed_effects.do	6	name, lat, loc, location
Code	B_gender_moves.do	7	gender, lon, loc, location
Code	B_summary_stats.do	11	name, lat
Code	README.tex	150	url, lon, lat, son, loc, location, name, dob, second, gender, city, country, minute, social, email, census, zip
Code	calibrate_NAM.py	1	lat
Code	ebayes.ado	52	name, loc, lat, second, degree, lon
Code	ebayes.ado	52	name, loc, lat, second, degree, lon
Code	fese_fast.ado	18	lat, city, name, loc
Code	fese_fast.ado	18	lat, city, name, loc
Code	run_stata_codes.do	1	gender
Code	selection_based_NAM.do	6	lat

File Type	File	Variables/References	PII Categories
Code	<code>simulate_selection.py</code>	7	name, lat, loc

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-05-18T11:01:42.258

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (<code>C:\</code> etc)
25	1	9	1	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/Replication/CodeB/Stata codes/fese_fast.ado	0.0	e18776996ac50b147dbd51ba774dc0eb8fe2da6f
/Replication/CodeB/Stata codes/ebayes.ado	0.01	295ddb30a99b2a2c61ecc67819429559ff3444c6

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

- Folder A contains 2 R scripts, a master Stata do-file which references to 5 seperate stata files. Additionally it contains 2 python files and another separate stata do-file for the simulation part of the study. Folder B replicates the same structure of folder A for the analysis for company B instead of company A, excluding the simulation.
- It is unclear from the ReadMe which in-text numbers are replicated where. It would be very helpful if the authors provided more details. For instance, “the numbers cited in the related discussion in the text” is not precise enough for the replicator to associate the code with specific numbers in the text.

cloc

github.com/AIDanial/cloc v 2.02 T=0.05 s (508.2 files/s, 139599.5 lines/s)

Language	files	blank	comment	code
Stata	16	1047	275	2570
TeX	1	208	5	1318
R	4	183	80	336
Python	2	77	48	171
---	---	---	---	---
SUM:	23	1515	408	4395

☐ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

{SUGGESTED} We strongly advise the use of a single (or a small number of) main control file(s) to automatically reproduce all figures and tables in the paper, without manual interaction.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Nuvolos
- R 4.4.0
- Stata/MP 19.5 (version 19 was not available to the replicator on nuvolos.)
- JupyterLab 4.1.6 with Python 3.11

8 Replication

- Downloaded code and data from dropbox.
- Committed code to git repo.
- Installed all R packages mentioned in the ReadMe.
- Installed all Stata packages mentioned in the ReadMe.
- Imported all Python libraries as per ReadMe.
- Running R script A_estimate_fe.R in CodeA as per ReadMe resulted in the following error message:

```
library(haven)
```

```
Error: package or namespace load failed for 'haven' in dyn.load(file, DLLpath = DLLpath,
unable to load shared object '/usr/local/lib/R/site-library/haven/libs/haven.so':
/usr/local/lib/R/site-library/haven/libs/haven.so: undefined symbol: R_getVar
```

```
In addition: Warning message:
```

```
package 'haven' was built under R version 4.5.0
```

- I switched to R version 4.5.2, even if the authors mentioned R.4.4.0 in the ReadMe. Now the 2 R scripts ran through.
- Moving on to the STATA code, after adjusting the path file in the stata master file, the code broke due to the require package not being listed in the required packages. After installing it through SSC the code ran to completion.
- To run the selection_based_NAM.do file I had to adjust the paths in the following manner, as the folder structure is not updated to suit the replication package:

```
global path "/files/JPE-Sollaci-20240098/replication-package/Replication"
```

```
gl data_dir "$path/Data"
```

```
gl code_dir "$path/CodeA"
```

```
gl res_dir "$path/Results"
```

- Moving to the Python scripts which I run in JupyterLab, the calibrate_NAM.py file runs to completion. The simulate_selection.py file broke because I needed to specify the path in the following fashion, where-after the code ran to completion.

```
## Set directory
```

```
os.chdir("/files/JPE-Sollaci-20240098/replication-package/Replication/CodeA/NAM Simulatio
etc.
```

- Next, I move on to company B, in the CodeB folder. The R scripts run to completion. The master do file for the stata code ran to completion.

9 Findings

9.1 Missing Requirements

- ☐ Software Requirements
 - ☒ Stata
 - package: require (was missing from the required packages list)
 - ☐ R
 - ☐ Version mentioned in ReadMe (4.4.0) was not compatible with the haven package.
 - ☐ Python
 - ☐ Version (missing)
- ☐ Computational Requirements specified as follows:
 - Cluster size, disk size, memory size, etc. (not specified)
- ☐ Time Requirements
 - Length of necessary computation (hours, weeks, etc.) (not specified)

[REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

A_estimate_fe.R prepares the following datasets: A_connected.dta, A_manager_fe.dta, A_store_fe.dta.

A_estimate_gfe.R prepares the following datasets: A_manager_gid.dta, A_manager_gfe.dta, A_store_gid.dta, A_store_gfe.dta.

A_fixed_effects.do prepares A_dataset_fe_eb.dta.

A_grouped_fixed_effects.do prepares A_dataset_gfe.dta.

A_event_study.do prepares A_event_study.dta.

B_estimate_fe.R prepares the following datasets: B_connected.dta, B_manager_fe.dta, B_store_fe.dta.

B_estimate_gfe.R prepares the following datasets: B_manager_gid.dta, B_manager_gfe.dta, B_store_gid.dta, B_store_gfe.dta.

B_fixed_effects.do prepares B_dataset_fe_eb.dta.

B_grouped_fixed_effects.do prepares B_dataset_gfe.dta.

B_event_study.do prepares B_event_study.dta.

selection_based_NAM.do prepares simulate_match.dta.

9.3 Tables and Figures

- Table 1: There are small deviations in the variance table. See below:

Company	Line	Column	Original Value	Reproduced
A	2	3	1.80	1.82
A	2	4	1.03	1.05
A	2	5	-1.16	-1.14
B	1	3	1.63	1.65
B	1	4	1.52	1.51
B	2	3	0.72	0.74
B	3	3	1.04	1.05
B	3	4	0.76	0.77
B	4	5	-0.03	-0.02

- Table 2: The numbers reproduced do not match the table in the manuscript. See screenshots below:

	manager...	store_bin	freq	lab_mana...	lab_store
1	1	1	33	Top	Top
2	2	1	49	Middle	Top
3	3	1	88	Bottom	Top
4	1	2	50	Top	Middle
5	2	2	66	Middle	Middle
6	3	2	79	Bottom	Middle
7	1	3	97	Top	Bottom
8	2	3	71	Middle	Bottom
9	3	3	61	Bottom	Bottom

Reproduced output should correspond to the company A side of table 2

	manager...	store_bin	freq	lab_mana...	lab_store
1	1	1	121	Top	Top
2	2	1	228	Middle	Top
3	3	1	463	Bottom	Top
4	1	2	219	Top	Middle
5	2	2	336	Middle	Middle
6	3	2	284	Bottom	Middle
7	1	3	458	Top	Bottom
8	2	3	323	Middle	Bottom
9	3	3	171	Bottom	Bottom

Reproduced output should correspond to the company B side of table 2

Table 2: Number of manager and store matches

		Stores					
		Company A			Company B		
		Top	Middle	Bottom	Top	Middle	Bottom
Managers	Top	33	51	97	116	229	461
	Middle	49	67	69	225	332	285
	Bottom	88	76	64	457	322	173

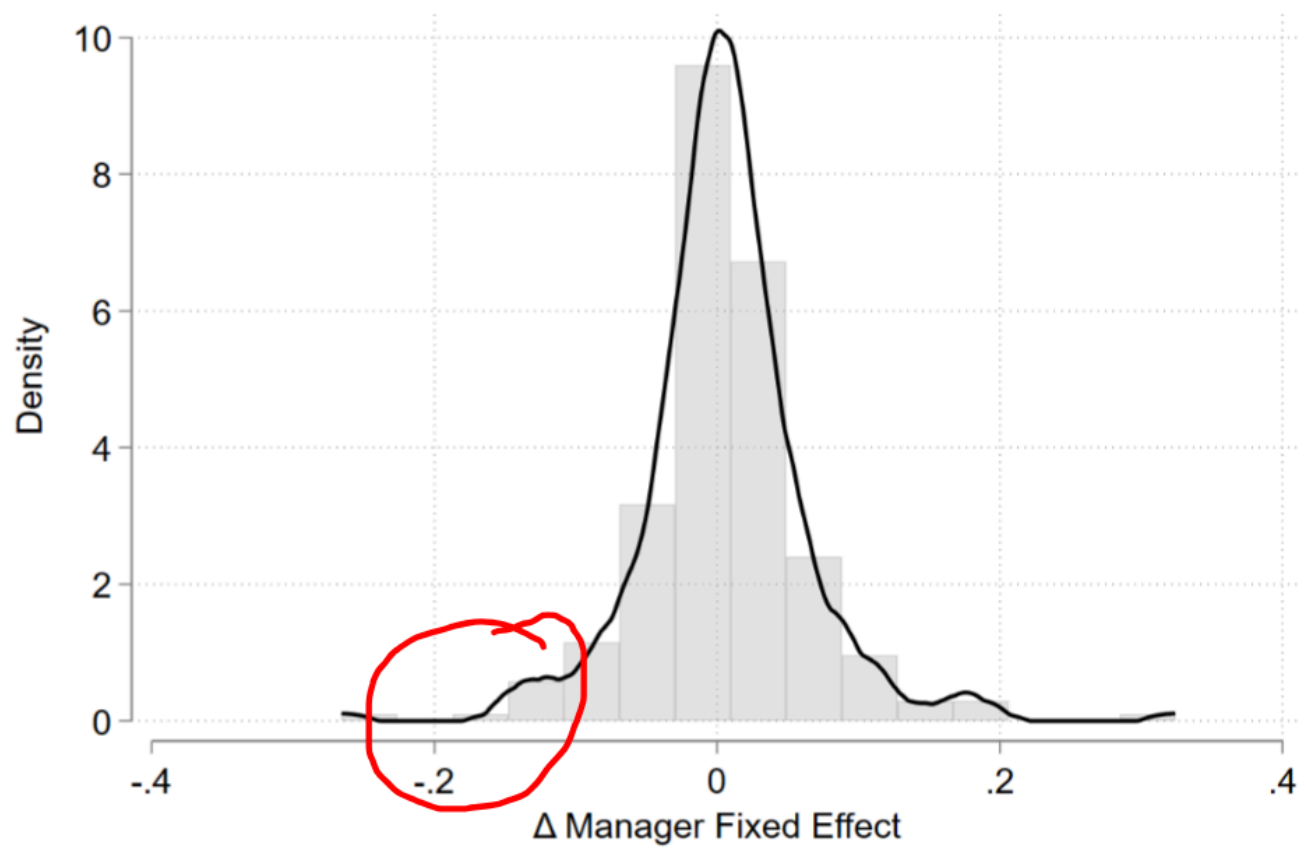
Paper version

Figure 1

- Figure 2: looks the same.
- Figure 3: There are 4 figures that are called figure 3a, figure 3b, etc. It’s not clear to the replicator which figure corresponds to which. For instance, the authors might consider creating a table in the ReadMe which maps the outputs to the corresponding figures. This would greatly reduce the time burden for the replicator.
- Figure 4: Looks the same.
- Figure 5: looks the same (mostly), but in the paper full-time employment and full-time/part-time ratio are being confused (there are two figures with the ratio in the y-axis), but the titles are different. The replicator can’t attribute which of the figures is correct and which isn’t.
- Figure 6: looks the same.
- Figure 7: Reproduced, but with differences:



Reproduced output for a)

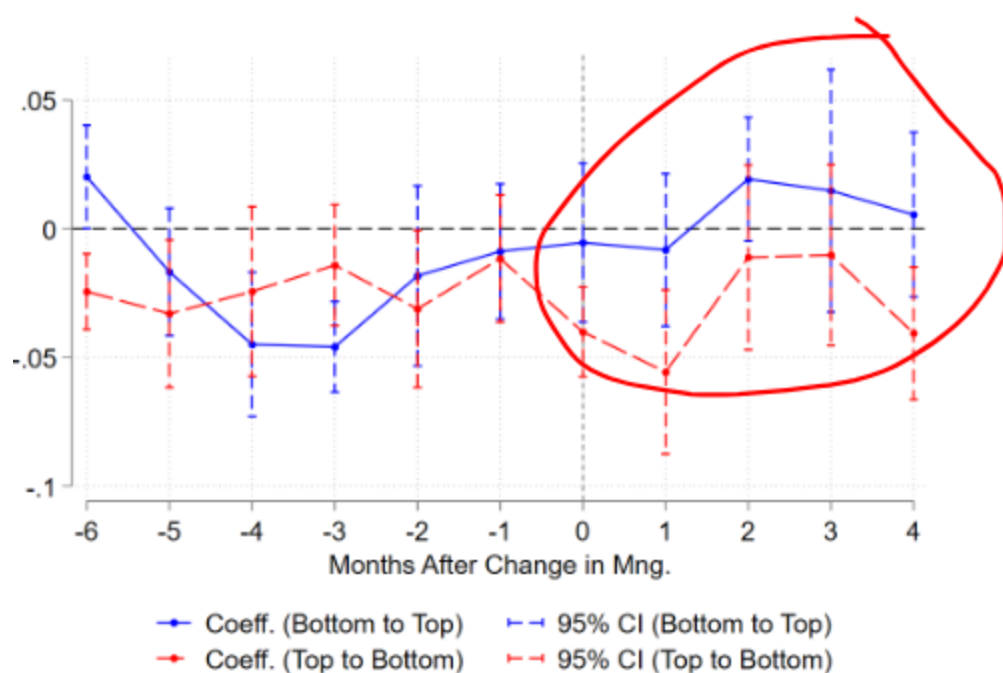
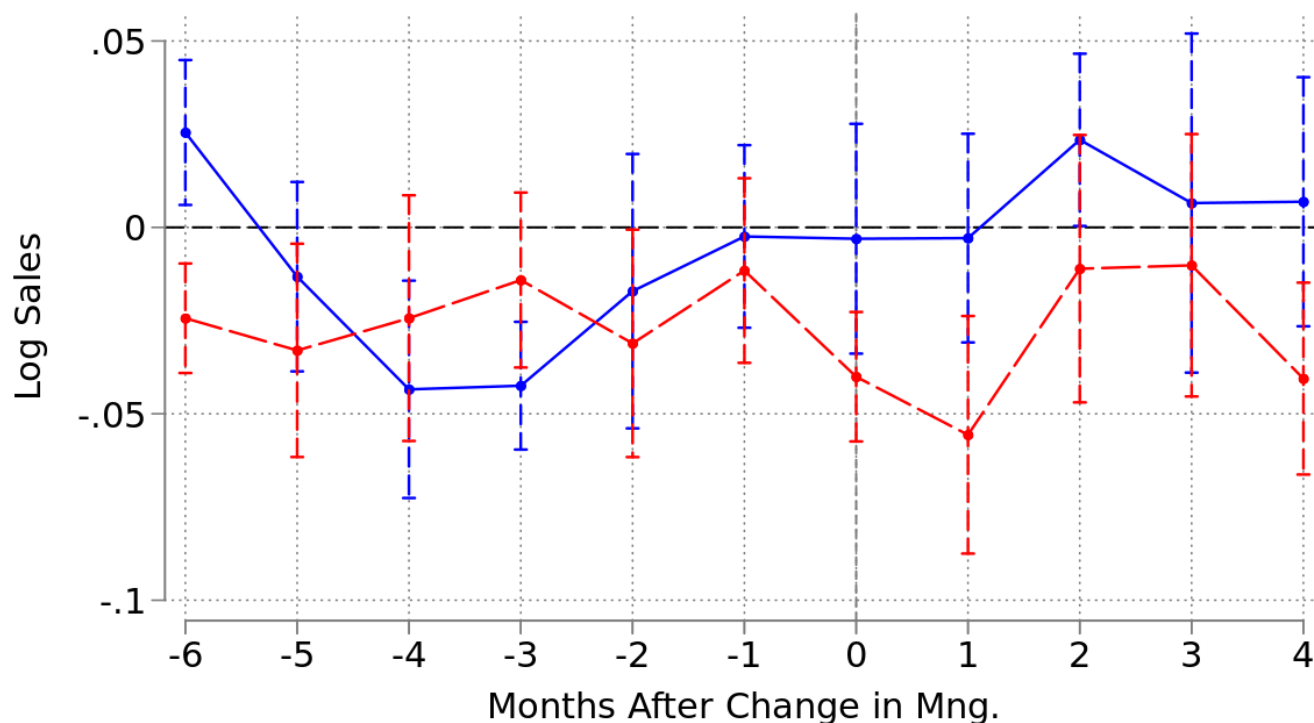


(a) Manager FE (Company A)

Paper version

Figure 2

- Table 3: Numbers diverge similar as table 2.
- Figure 8: Some of the event graphs look different. I included b) as an example below.



(b) Sales (log)

- Figure 9: Some of the event graphs display discrepancies similar to figure 8.
- Table 4 was not saved in the results folder. The results displayed in stata were diverging as well, for example company B After Change Standard Deviation reproduced displayed 0.066 while the paper

reads 0.64 for that number.

- Figure A1 looks the same.
- Figure A2 is showing discrepancies:

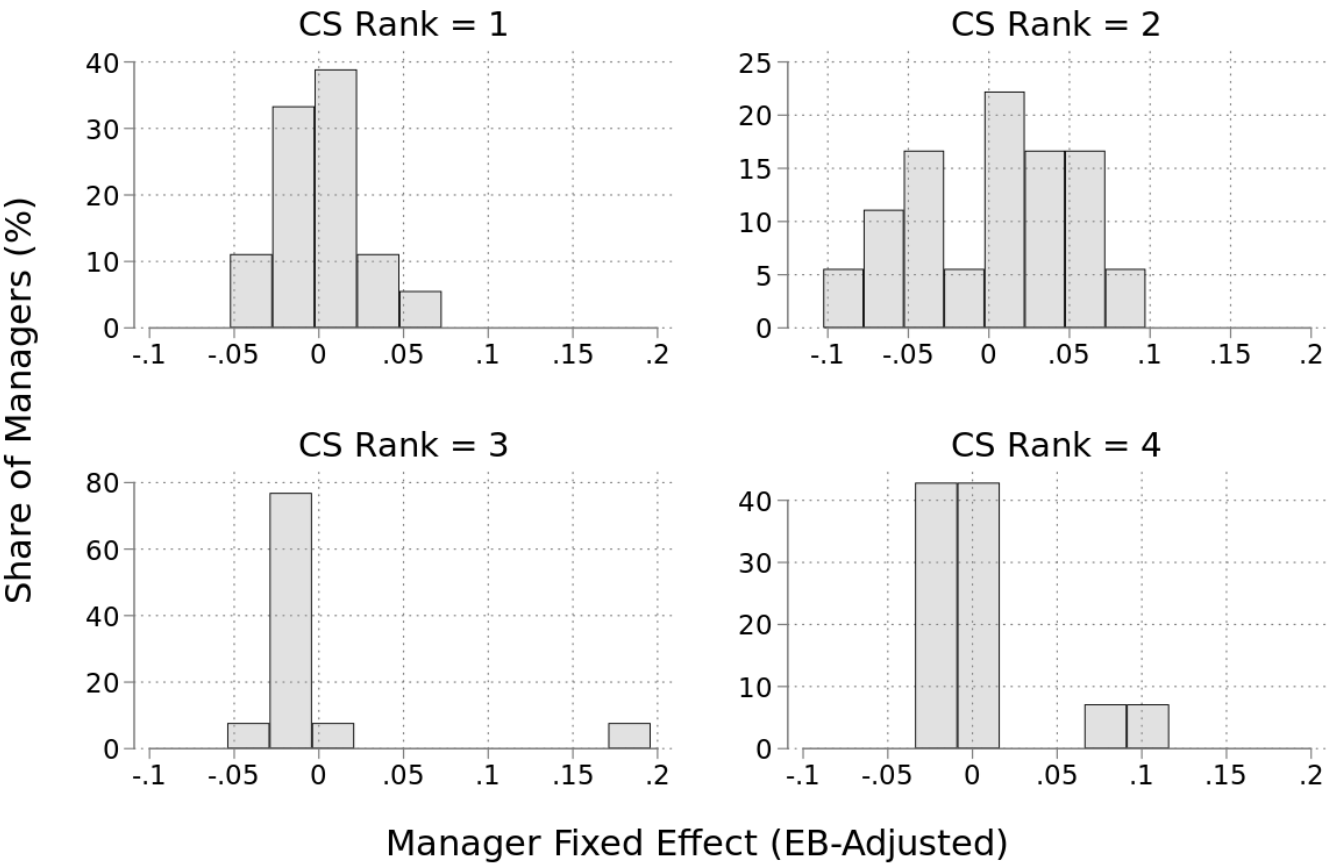
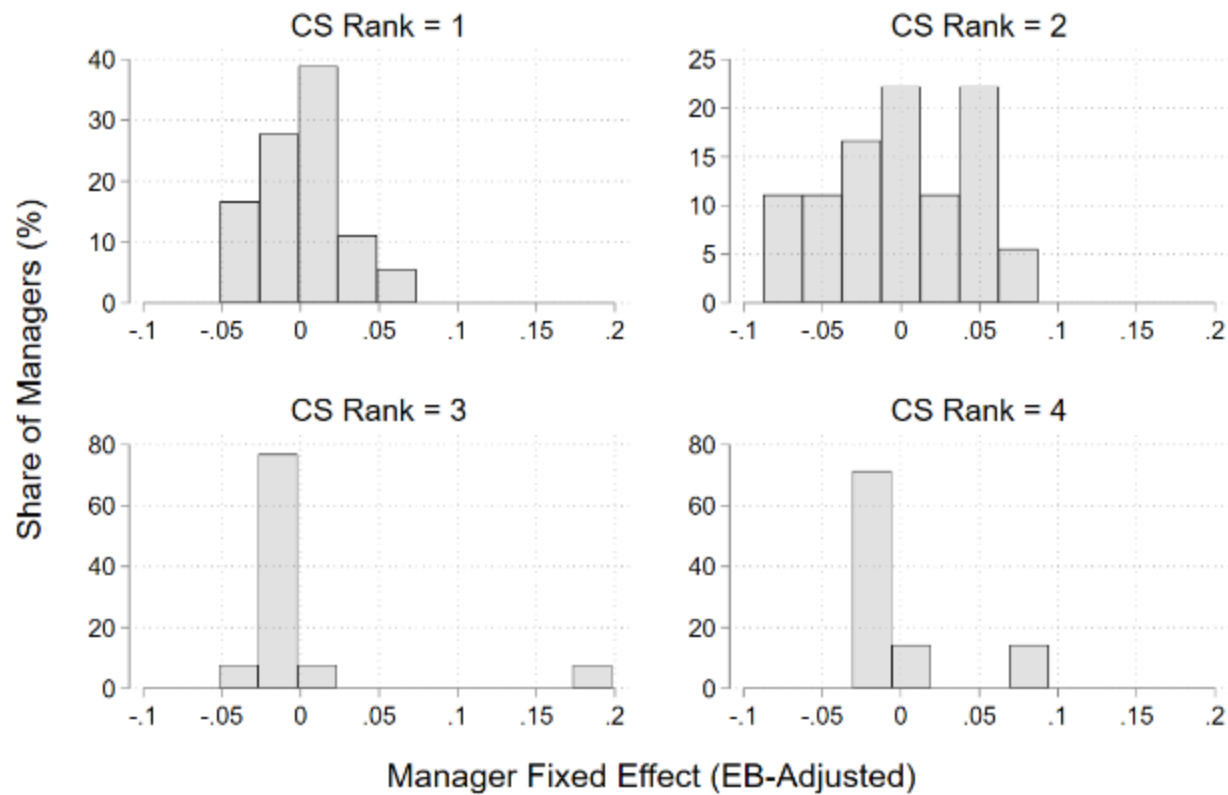
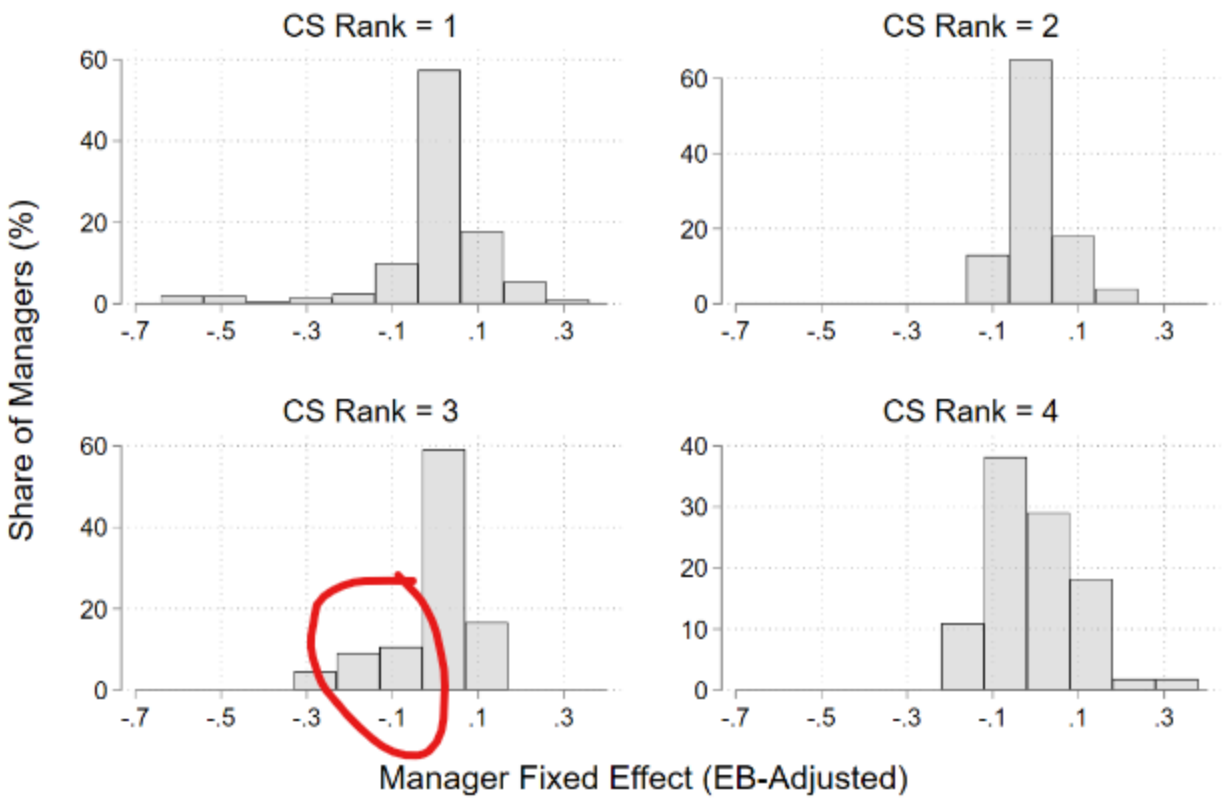
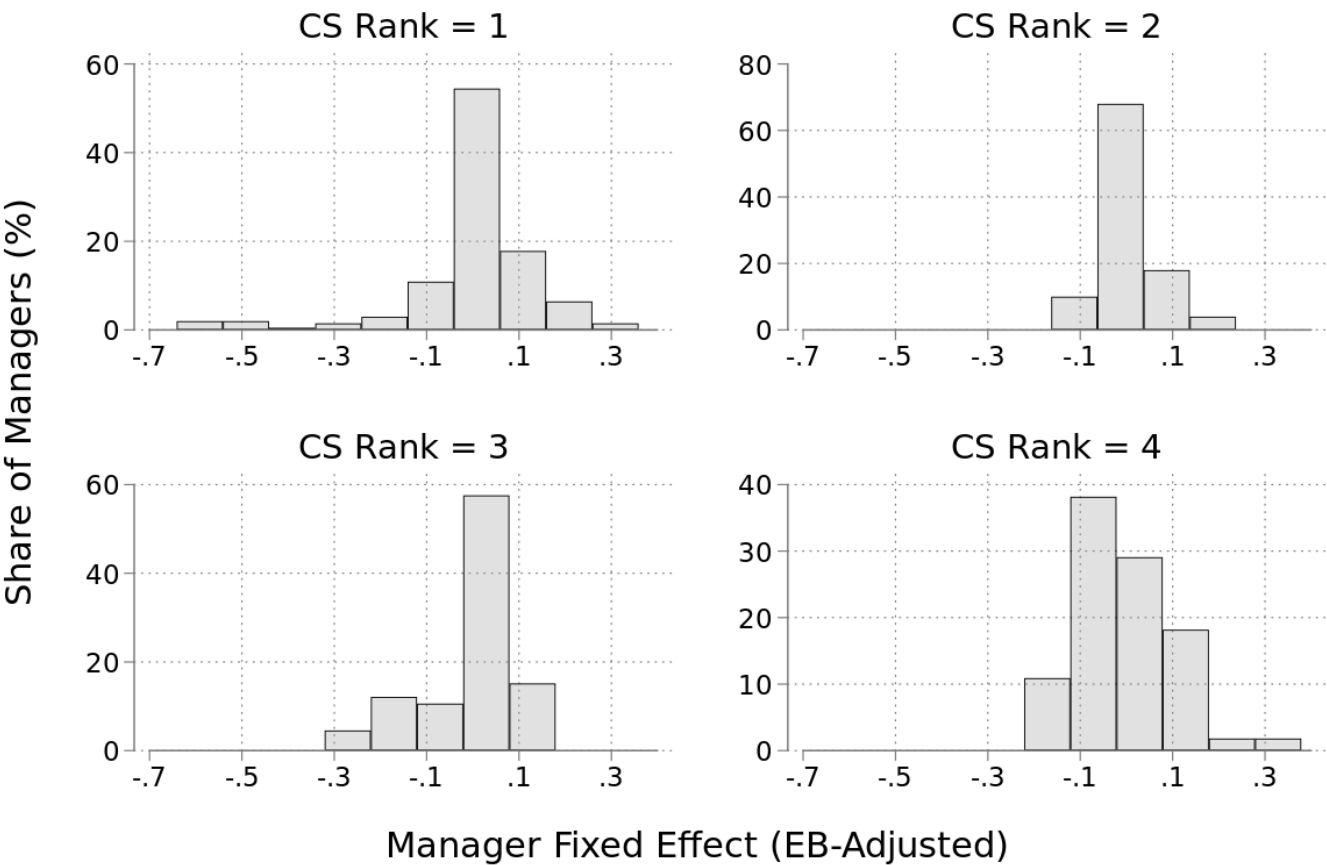


Figure A.2: Distribution of EB-Adjusted Manager Fixed Effects



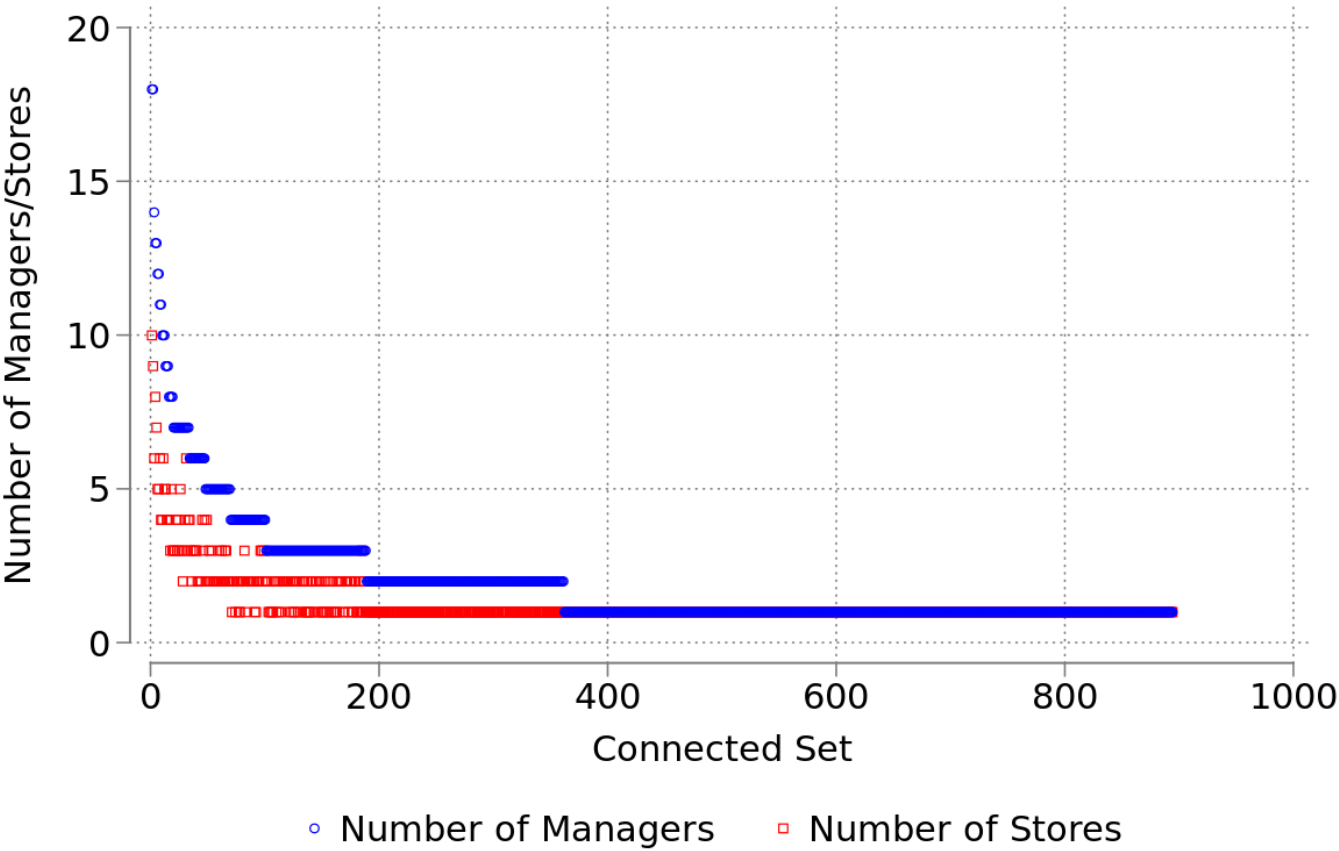
(a) Company A

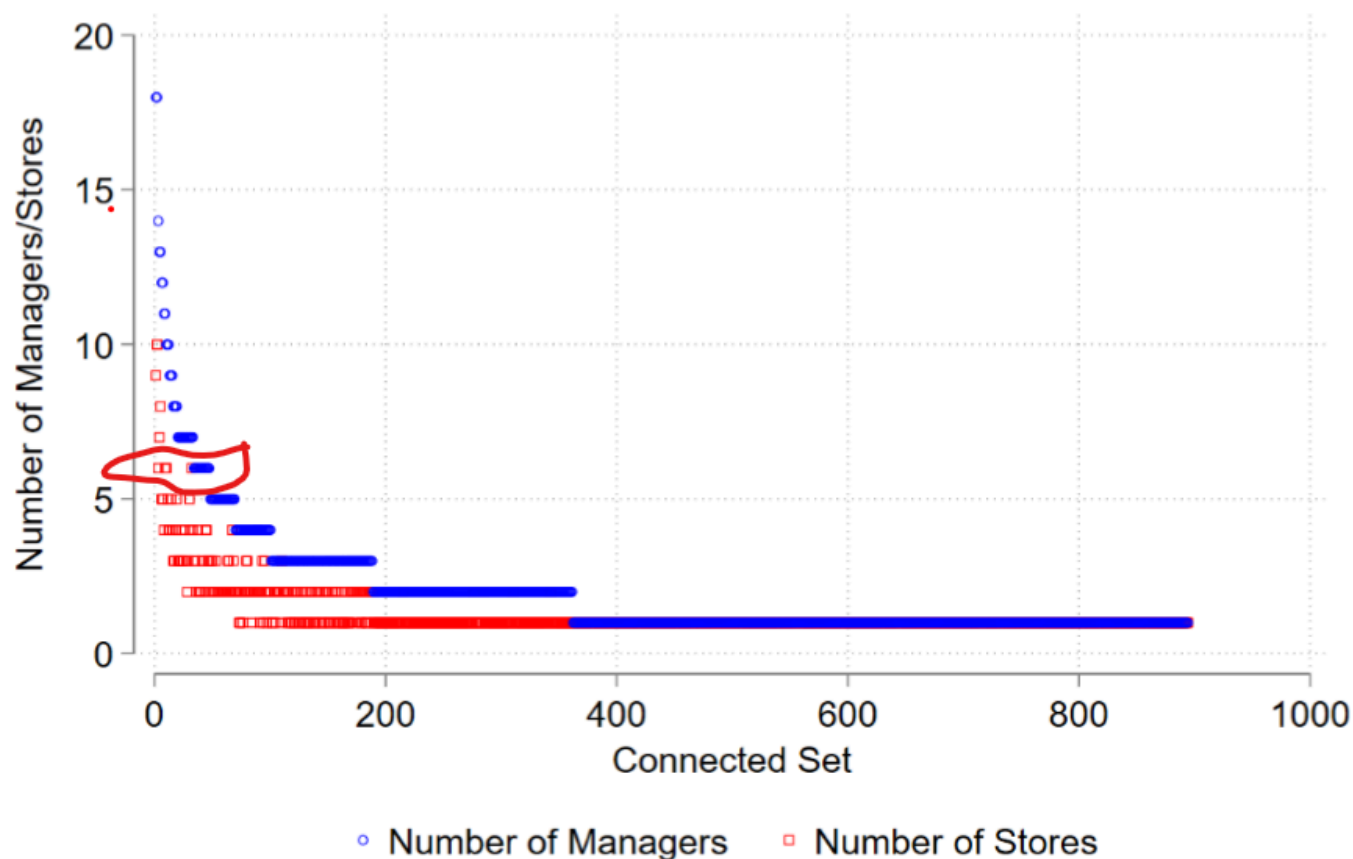


(b) Company B

Note: FE's de-meaned by connected set for plotting. 46

- Figure A3 showcases similar discrepancies as figure A2.
- Figure A4 Company A showcases discrepancies:





(a) Company A

- Figure A5: Looks the same.
- Table B.1 was not saved in the results folder. The numbers were displayed in the stata console.
- Table B.2 was not saved in the results folder. The numbers were displayed in the stata console.
- Table C.1 was not saved in the results folder. The numbers were displayed in the stata console.
- Figure C.1 looks the same.
- Figure C.2 looks the same.
- Figure C.3 looks the same.
- Table D1 was not saved in the results folder. The numbers were displayed in the stata console, company B has a diverging coefficient -0.211 (reproduced) vs -0.212 (manuscript).
- Figure D1 displays discrepancies for both companies:

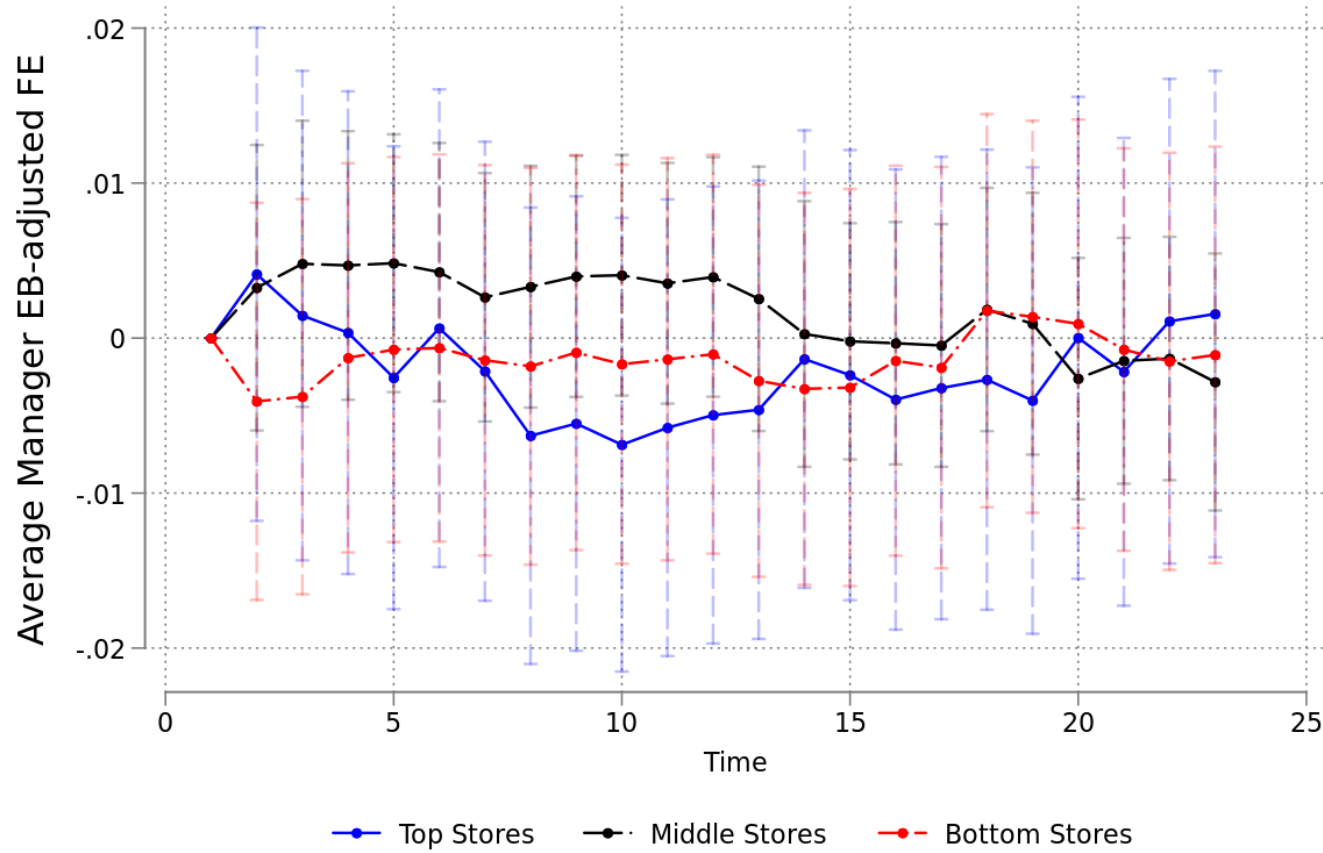
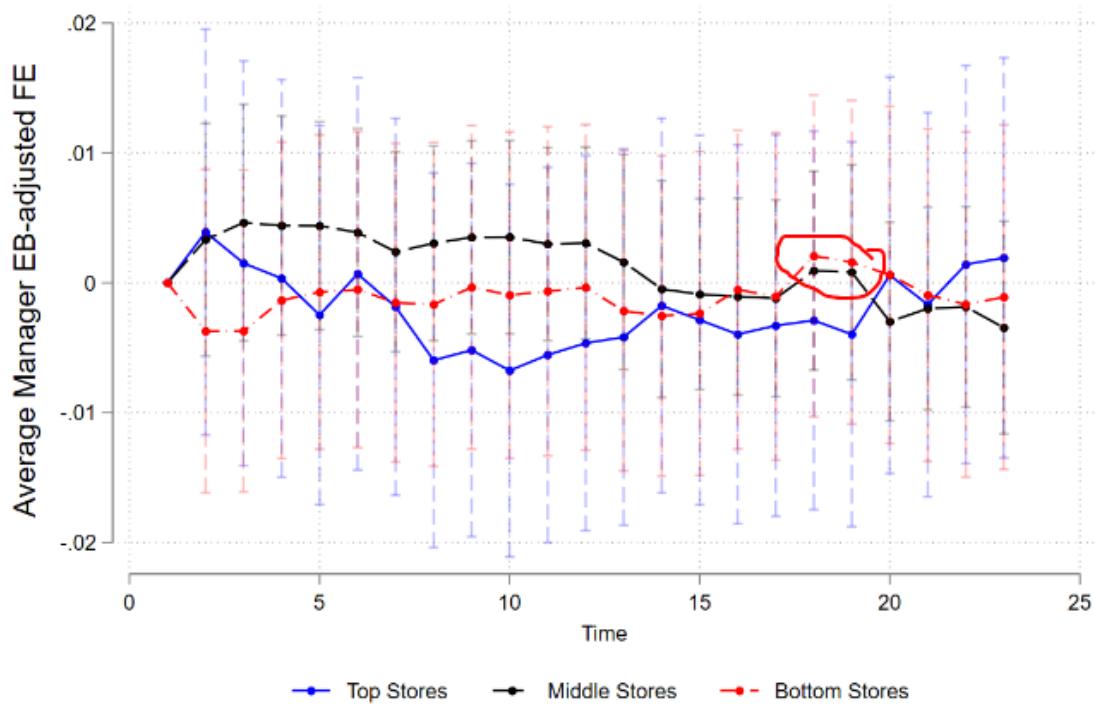
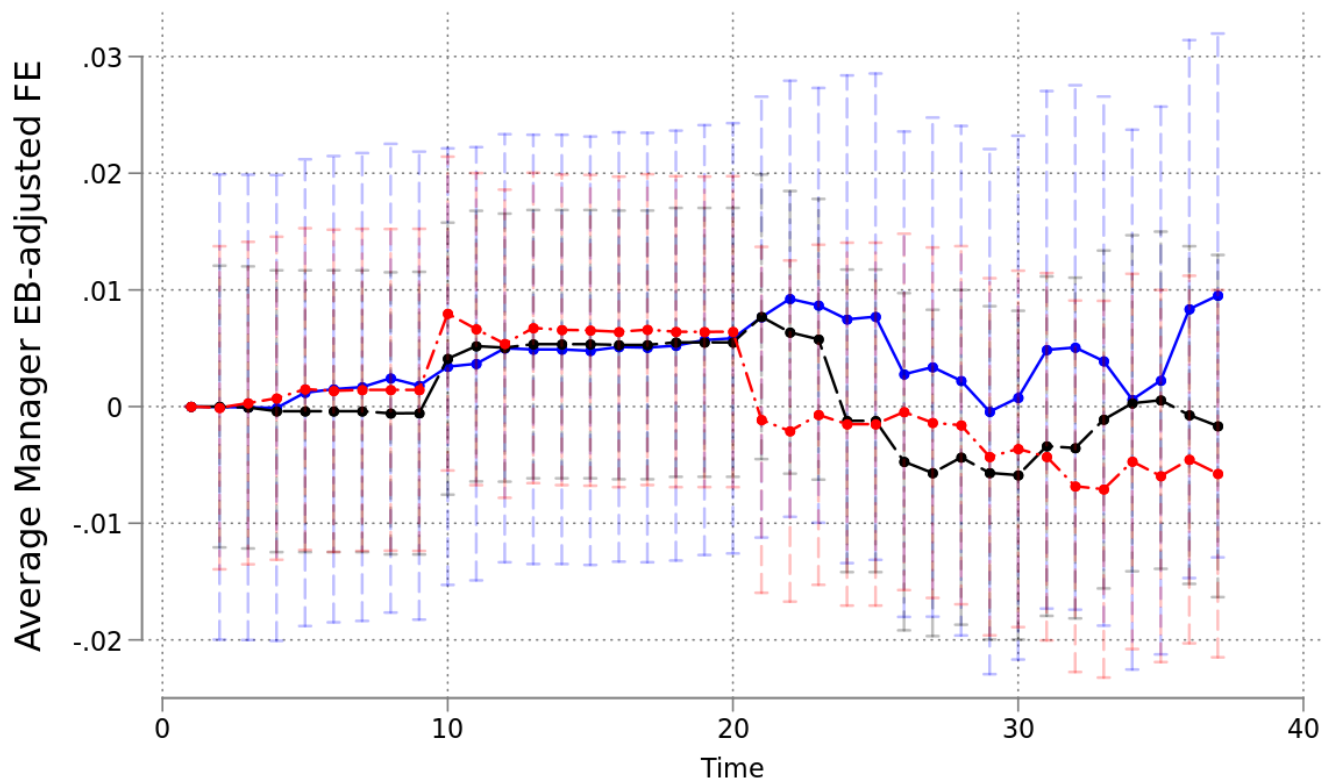


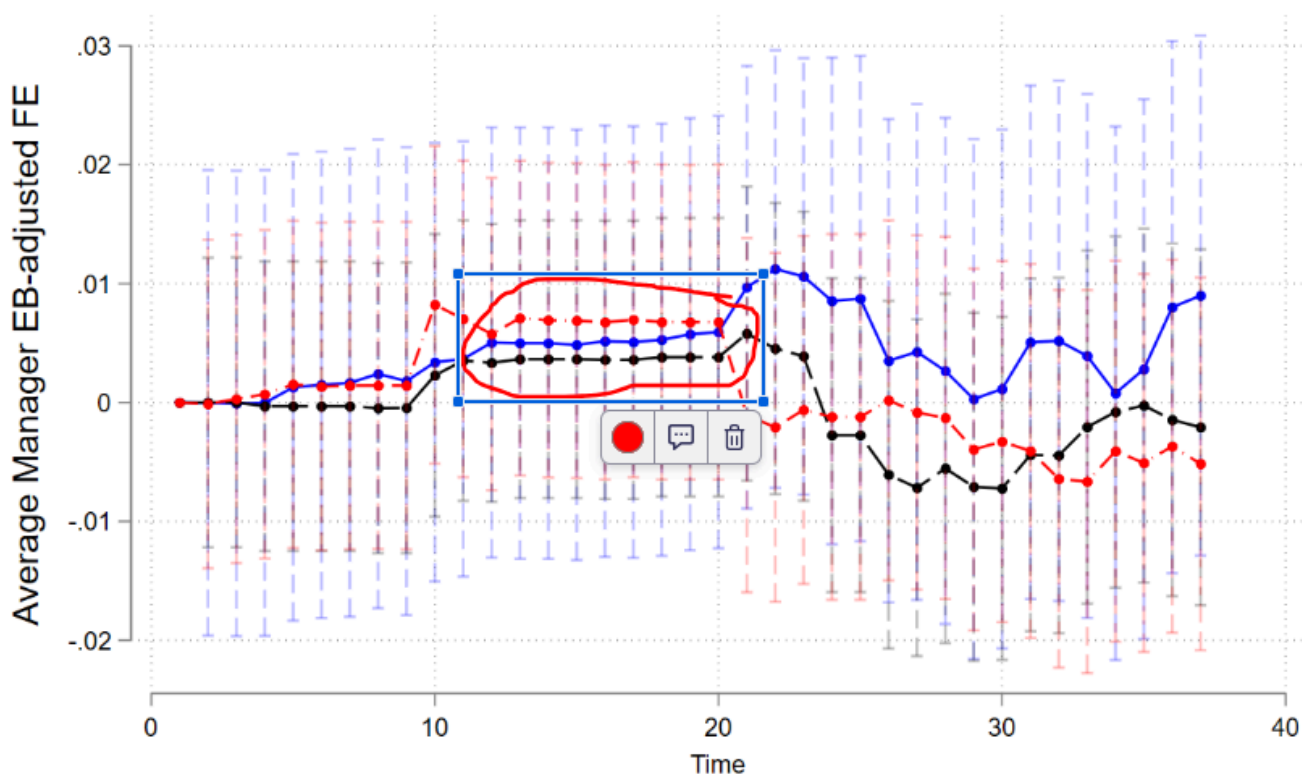
Figure D.1: Evolution of the Average Manager FE by Store Quality Tercile



(a) Company A



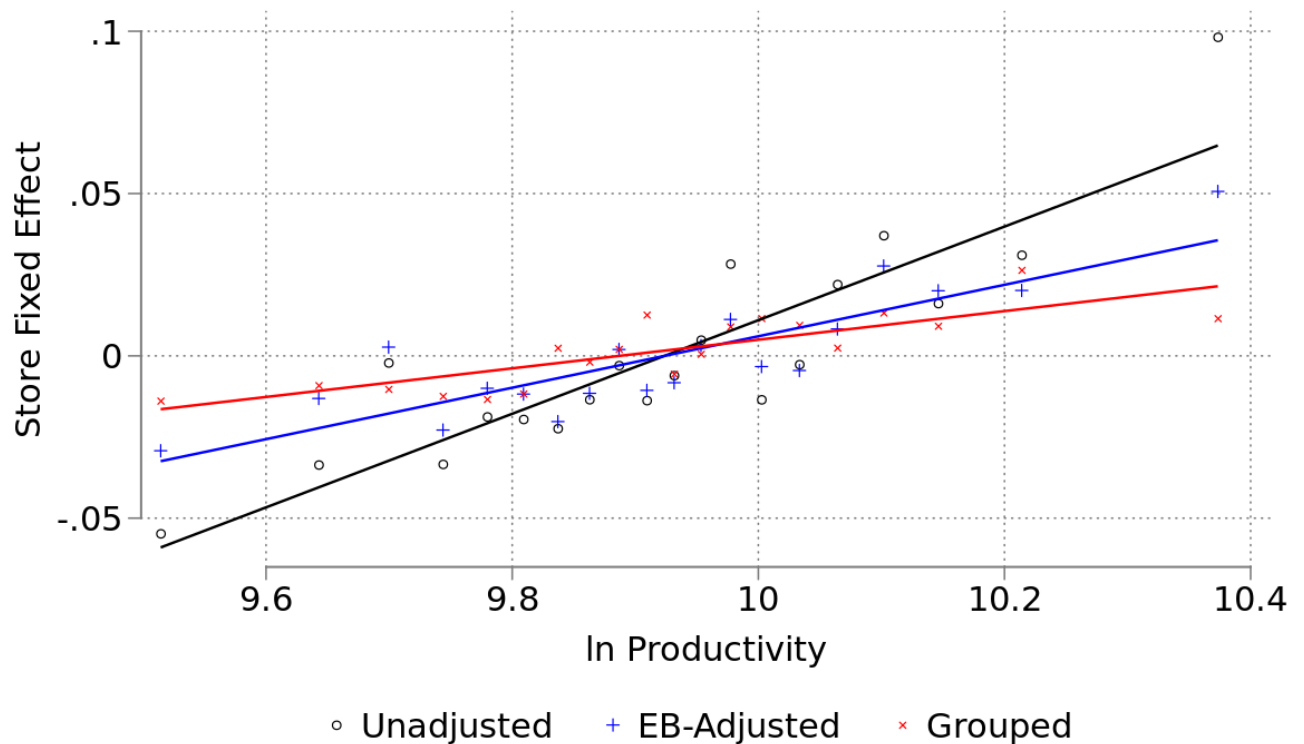
Top Stores Middle Stores Bottom Stores



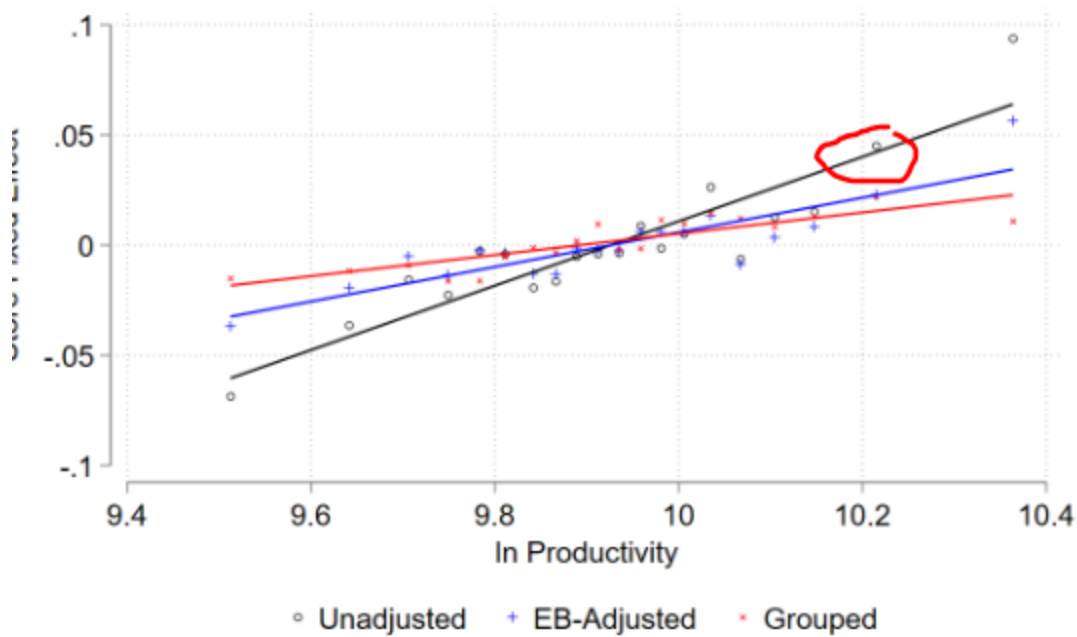
Top Stores Middle Stores Bottom Stores

(b) Company B

- Figure D2 looks the same.
- Figure D3 showcases some discrepancies, but the figures are based on a randomised simulation, which could explain reasonably minor differences:
- Figure E1: looks the same.
- Figure E2: looks the same.
- Figure E3 was reproduced, but showcases discrepancies.

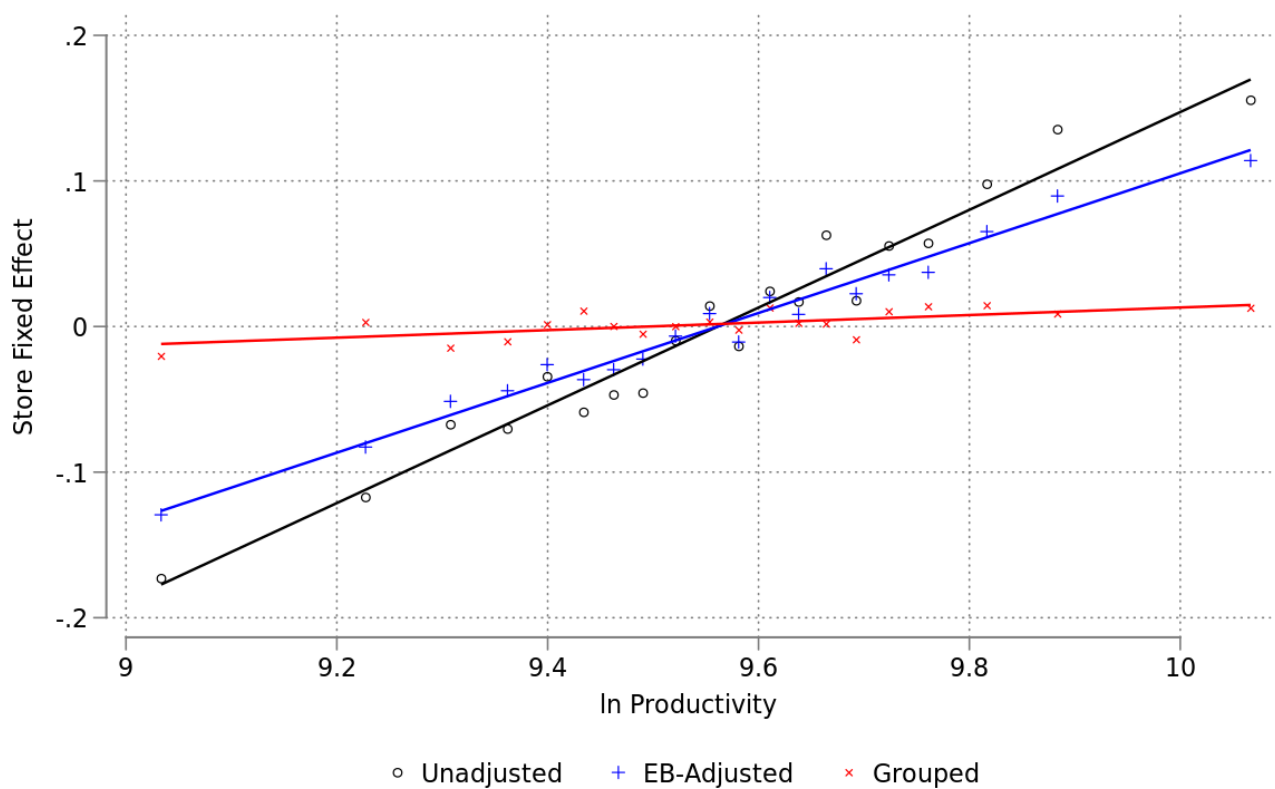


Note: all fixed effects are normalized so that their average is zero within the connected set.

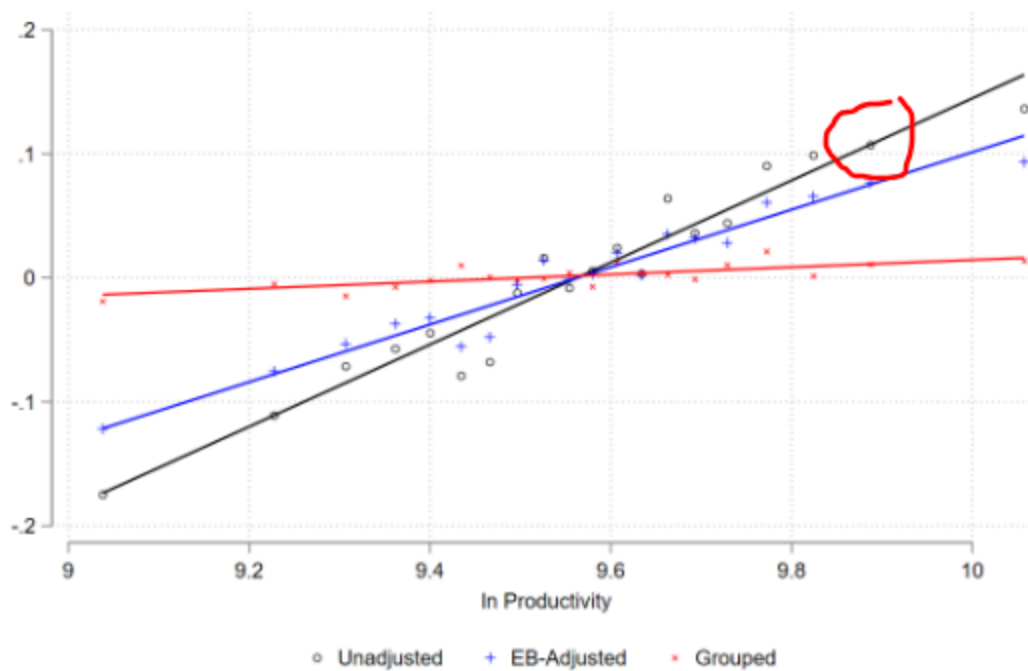


Note: all fixed effects are normalized so that their average is zero within each connected set.

(b) Store Fixed Effect (Company A)



Note: all fixed effects are normalized so that their average is zero within each connected set.

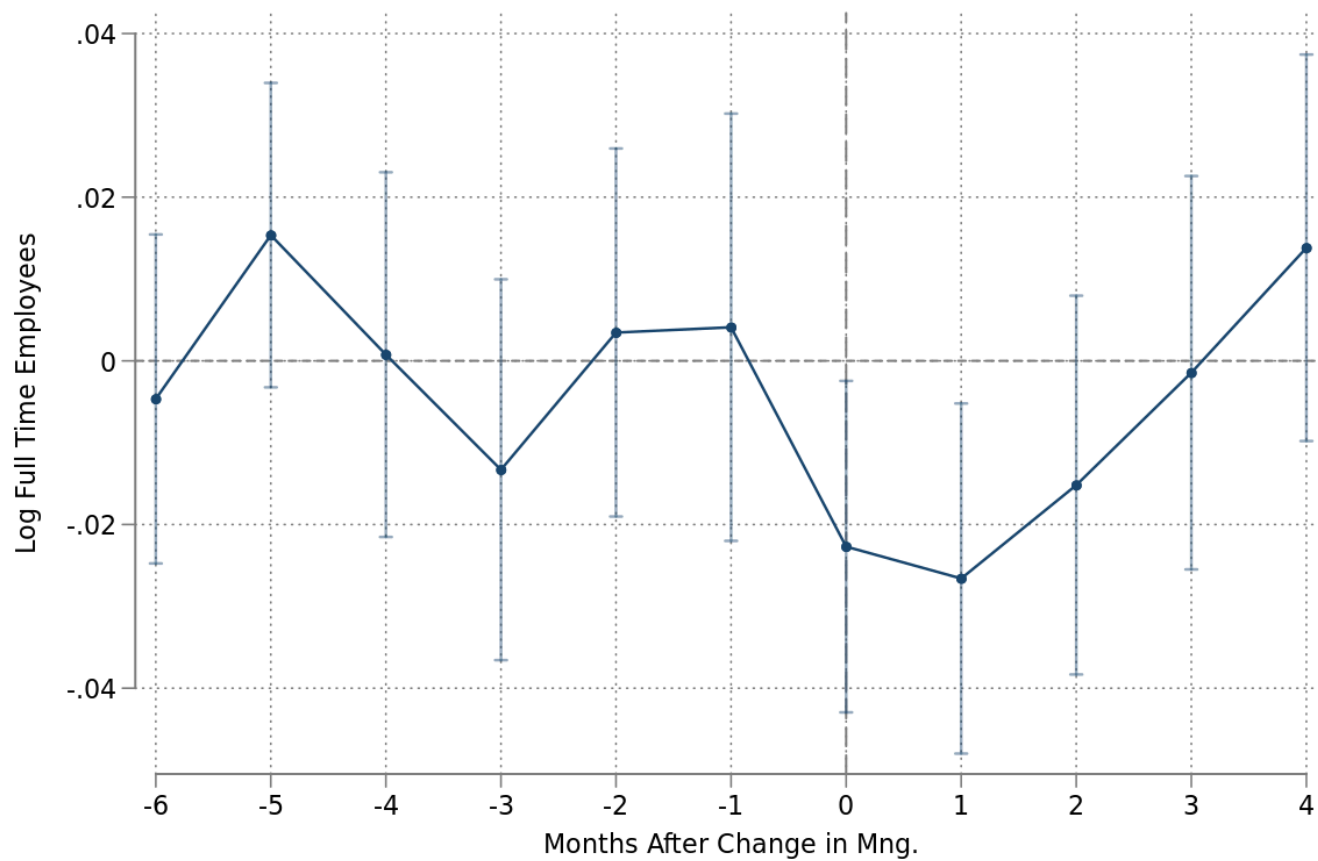


Note: all fixed effects are normalized so that their average is zero within each connected set.

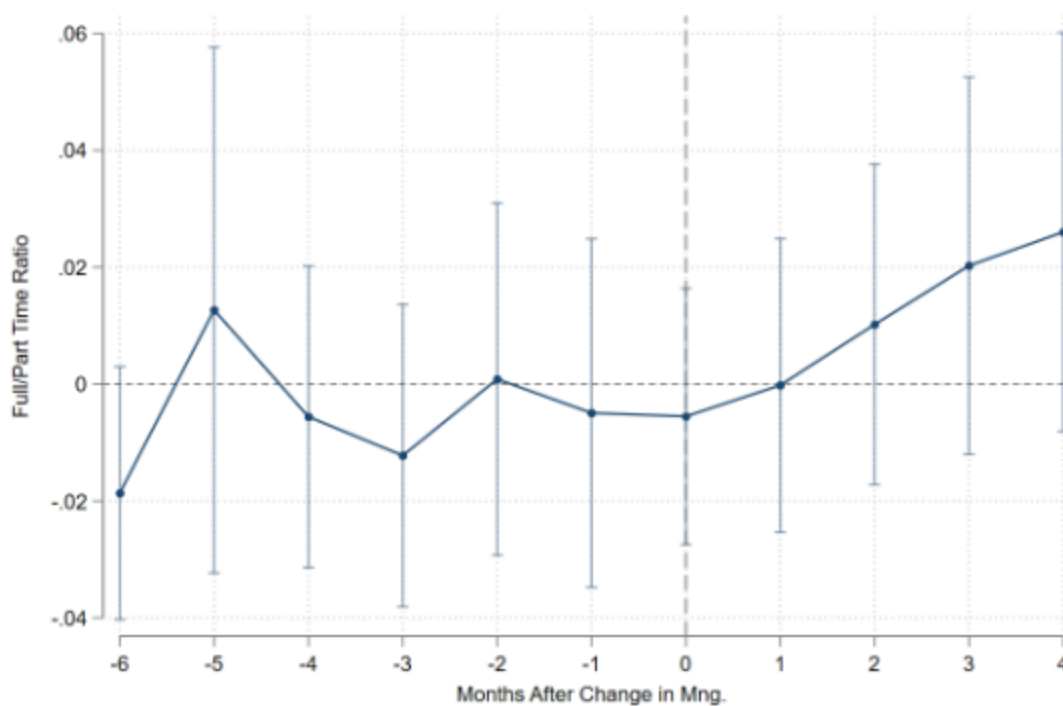
(d) Store Fixed Effect (Company B)

Manuscript figure E3 (d)

- Table F.1 was not saved in the results folder, the number displayed in the stata console for Company A, percent estimate slightly diverges form the manuscript (0.66 vs 0.65)
- Figure G.1: looks the same.
- Figure G.2: looks the same.
- Figure G.3: discrepancy in (c) full-time employment.



Reproduced figure G3 (c)



(c) Full-Time Employment (log)

Manuscript figure G3 (c)

- Figure G.4: looks the same.
- Figure G.5: looks the same.
- Figure G.6: looks the same.

9.4 In-Text Numbers

{REPLICATOR}: list page and line number of in-text numbers. If ambiguous, cite the surrounding text, i.e., "the rate fell to 52% of all jobs: verified".

- p. 11 "Our data contains about 890 connected sets in Company A and just over 1,200 connected sets for Company B": verified
- p. 11 "The four largest connected sets in Company A have between 13 and 18 managers and 7 to 10 stores.": verified
- p. 12 "Company B. The largest set has 202 managers": verified "and 130 stores.": NOT verified, the replication exercise returns 131 stores.
- p.12 "Set sizes to [this reads like a typo?] drop quickly, however, with the next 3 largest connected sets respectively containing 100, 66, and 55 managers and 67, 44, and 32 stores.": verified.
- p.52 "In Company A the correlation between store fixed effects under the two methods is 0.98, and the correlation between manager fixed effects is 0.76. In Company B, these correlations are 0.93 and 0.80, respectively": Not verified. Did not see those numbers in the referenced code from the ReadMe.
- p. 72 "Averaging these ratios, we find that the mean of r90m stands at about 0.005, with a standard deviation of 0.112. Increasing the percentile to 99 changes the average ratio r99m to about 0.021, with a standard deviation of 0.568.": verified, but the code that displays these numbers is referenced wrongly as Appenix C.2 and not E.2.
- p. 31 "In Company A the tenure gap is 14 months and significantly different from zero at the 1% level. The gap is smaller in Company B, only 3.4 months and not statistically significant at the usual levels.": verified.
- p. 8 "60% change managers only once, 30% change managers twice, and the remainder more frequently than that. Similarly, 14 or 23% of managers (depending on the company) work in two different stores, and a small fraction move through three or more. Between 20 and 30% of each company's manager changes are internal": not verified, didn't find the numbers under the heading in the code indicated in the ReadMe.
- p. 73 "almost USD 700,000 each month": not verified, I found over 700 000 USD not under."

- p.73 "about GBP 4 million": verified.
- p.10 footnote 7 "Female managers are about 7% less likely to move to another store than male managers, a gap that cannot be explained by manager tenure or store characteristics": verified.

9.5 Other Issues

- Title of the Readme needs to be adjusted.
- Line of code where results are reproduced in the codes are not provided in the ReadMe.
- There is no clear mapping of tables/figures in the manuscripts to the tables and figure names in the results folder. Many tables and results are not saved in the results folder which makes the replication time consuming. The lines of codes where tables and figures are reproduced need to be checked in the code files.

10 Classification

- ☐ full reproduction
- ☒ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☒ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package

- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements